Segmentation of Organoids in Grayscale Tomographic Images : A Comparative Approach Using Intensity-based Thresholding and Deep Learning Techniques"

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Abstract— This project explores various image processing techniques to improve the segmentation of organoids in grayscale microscopy images. Our objective is to isolate organoids from background noise and assess the performance of classical methods against a state-of-the-art deep learning model, MedSAM. We explore thresholding, edge detection, and normalization techniques, culminating in a hybrid pipeline to fine-tune MedSAM's output. Our experiments, evaluated via F1-score, suggest that classical preprocessing methods can meaningfully improve segmentation accuracy and even complement deep learning approaches.

This work is related to my github : https://github.com/Neil6mail/organoid-seeker

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I. Introduction

Organoids—three-dimensional cellular structures that mimic the architecture and function of real tissues—have emerged as valuable models in biomedical research. Accurate segmentation of organoids from radiographic images is crucial for downstream tasks such as size quantification, morphological analysis, and longitudinal tracking of development or treatment responses. However, this task is often complicated by various factors inherent in biological imaging: low contrast between foreground and background, inhomogeneous lighting conditions, partial occlusions, and the presence of imaging noise or artifacts. Consequently, achieving reliable and reproducible segmentation requires sophisticated processing techniques that go beyond simple thresholding or binary operations.

In this project, we explore the intersection of classical image processing techniques to develop a robust and efficient pipeline for organoid segmentation. Our work is motivated by a dual hypothesis: first, that careful statistical analysis and preprocessing of image intensity distributions can lead to meaningful segmentation in many cases without requiring machine learning; and second, that these classical methods can act as a strong complement to deep learning models such as MedSAM (Medical Segment Anything Model), enhancing their performance on noisy, ambiguous, or underexposed samples.

To this end, we investigated and compared several segmentation strategies. We began by studying the statistical structure of grayscale intensity values in order to separate background noise from biological structures. This led us to implement thresholding approaches, notably the Yen threshold, which leverages entropy-based criteria to determine optimal pixel value cutoffs. Next, we considered edge-based methods, particularly the Canny edge detector, and complemented it with shape-based filters such as elliptic fitting to identify potential organoids based on morphological criteria like eccentricity and area. Interestingly, we also explored unconventional strategies such as z-score normalization and gamma correction—typically used in other domains such as contrast enhancement or radiomics—which proved surprisingly effective in improving signal-to-noise ratio for downstream segmentation.

The second major focus of this project was the integration of MedSAM, a transformer-based segmentation model designed for medical imagery, into our pipeline. While MedSAM performs reasonably well out of the box, we hypothesized that its predictions could be further improved by feeding it preprocessed images. To test this, we compared raw MedSAM output to its output on images that had undergone various classical preprocessing routines. Evaluation was done using the F1-score, a standard metric balancing precision and recall, to assess segmentation quality relative to ground truth masks.

Our results demonstrate that image preprocessing is not merely a helpful auxiliary step but can serve as a foundational layer that directly contributes to better generalization and accuracy of segmentation models. These findings are particularly valuable in scenarios where annotated data is scarce, where model fine-tuning is computationally infeasible, or where interpretability and transparency are required—conditions commonly encountered in biological laboratories and clinical research settings.

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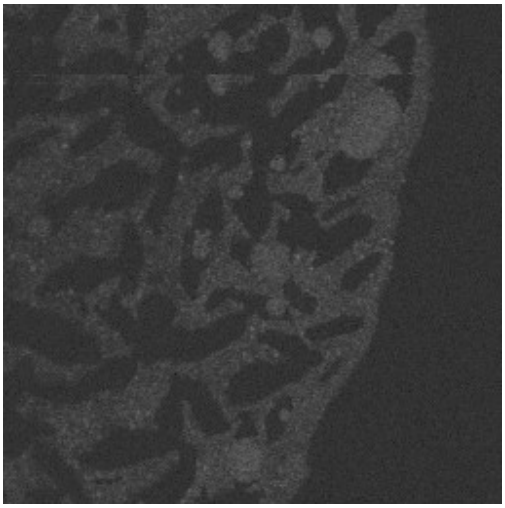

Fig 1. 2D slice of Tomography image

II. Background / Related Work

The problem of image segmentation—assigning labels to pixels in an image to isolate regions of interest—has been a foundational task in computer vision and biomedical image analysis. In the context of microscopy images, particularly grayscale radiographs used in cell biology, segmentation is often complicated by factors such as poor contrast, complex object shapes, overlapping structures, and inconsistent illumination. As such, a wide range of both classical and modern techniques have been proposed to address these challenges.

A. Classical Image Processing in Biomedical Imaging

Before the advent of deep learning, classical image processing methods were the dominant approach for segmenting biological structures. These methods typically rely on assumptions about intensity distributions, spatial continuity, and object morphology.

One of the earliest and most widely used techniques is thresholding, where pixels are classified based on their grayscale intensity values. Among the many thresholding strategies, Otsu's method [1] is notable for maximizing inter-class variance between foreground and background, while Yen's method [2] uses entropy-based optimization to capture more complex distributions. These approaches are particularly effective in images with bimodal histograms, which is often the case in clean microscopy environments.

Edge detection is another classical strategy, frequently applied to identify boundaries of biological structures. The Canny edge detector [3], introduced in 1986, remains one of the most effective due to its use of gradient filtering, non-maximum suppression, and edge tracking via hysteresis. In biomedical contexts, however, raw edge maps often contain excessive detail or noise. Therefore, additional filtering steps—such as shape-based constraints or morphological operations—are often required to retain biologically meaningful edges. For example, elliptic shape fitting has been used in prior work [4] to identify rounded organ-like structures by evaluating contour eccentricity and aspect ratios.

Some studies also highlight the importance of contrast enhancement techniques such as gamma correction and z-score normalization in preprocessing pipelines. While originally developed for display enhancement or statistical standardization, these techniques can substantially improve the visibility of low-contrast features when used as preprocessing steps prior to segmentation [5].

B. Deep Learning approach : From U-Net to MedSAM

Over the past decade, deep learning has revolutionized biomedical imaging, with models such as U-Net [6] and Mask R-CNN achieving state-of-the-art performance in tasks like cell segmentation, tumor detection, and organ delineation. U-Net, in particular, has become a standard model in the community due to its encoder–decoder architecture and skip connections, which help preserve spatial resolution and capture multiscale features.

More recently, transformer-based models have entered the field, bringing new capabilities in modeling long-range spatial dependencies. The Segment Anything Model (SAM), and its medical adaptation MedSAM [7], represent a new class of foundation models trained on massive amounts of labeled data and capable of zero-shot segmentation across a variety of imaging modalities. MedSAM has demonstrated strong performance on MRI, CT, and histopathology images. However, its performance on smaller datasets such as grayscale microscopy images can be inconsistent, especially when the input lacks strong contrast or clear boundaries.

Although deep learning offers unparalleled performance in many tasks, it also introduces several limitations. These models are data-hungry, often requiring large, well-annotated datasets for fine-tuning. They are also computationally intensive and prone to overfitting in the presence of noisy data or distribution shifts. Most importantly, they are black-box systems, offering limited interpretability—a significant drawback in biomedical applications where transparency and traceability are often critical.

C. Combining Classical and Modern Techniques

Given the respective strengths and weaknesses of classical and deep learning methods, recent research has begun exploring hybrid pipelines that leverage the best of both worlds. For instance, preprocessing with classical filters can clean and enhance input images before passing them to a neural network,

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often improving robustness and generalization [8]. Alternatively, classical methods can be used post hoc to refine or validate predictions made by AI models.

In the context of our project, we explore this complementary relationship by applying a set of preprocessing steps—including filtering, normalization, and contrast manipulation—before feeding the processed images into MedSAM. By simplifying the statistical structure of the image or enhancing key morphological features, these classical steps can help the model focus on relevant patterns and reduce confusion from background noise.

Our work contributes to this line of research by not only comparing classical and modern approaches side-by-side but also demonstrating how they can be synergistically combined to enhance the segmentation of organoids in grayscale radiographic data—an area where large datasets and high-quality annotations are typically unavailable.

III. Methodology

The task of segmenting organoids in grayscale tomographic images presents several unique challenges. These include low contrast between the cellular structures and the surrounding hydrogel environment, significant Gaussian noise across the dataset, and high morphological variability among organoids. These structures evolve over time—growing, changing shape, and sometimes overlapping—making their detection a nontrivial task. Our objective is to extract quantitative statistics from these evolving entities, which requires reliable segmentation across slices of a 3D tomographic reconstruction. *Authors and Affiliations*

Fig.2 : Ground-truth mask annotated by human

A. Dataset and Preprocessing

We work with a dataset composed of 500 grayscale radiographic slices, each capturing a 2D section of a biological sample. When stacked, these slices form a 3D tomographic representation of organoids suspended in hydrogel. Each 2D image is accompanied by a binary mask annotated by experts, highlighting the regions corresponding to organoids.

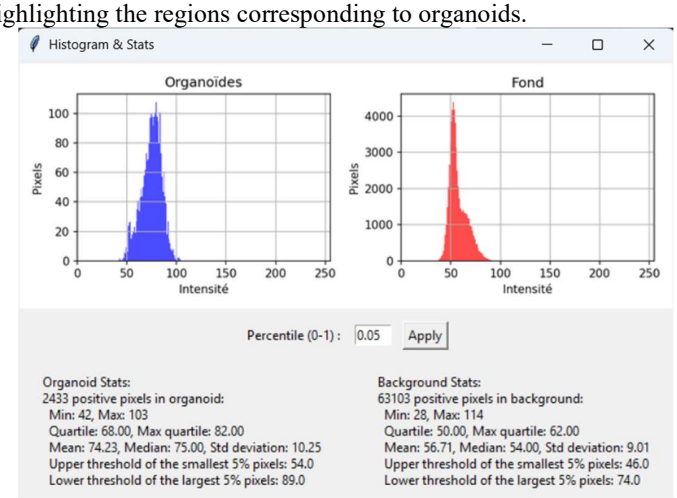

Fig.4 : Statistics on "organoides" pixel population and background population (named as Fond), showing the distribution of pixel's intensity in the grayscale image, from 0 to 255 as a value

To better understand the separability of organoids from the background based on intensity values, we first analyze the distribution of pixel intensities. To isolate statistics for the two regions (organoids and background), we perform bitwise masking operations. By applying a bitwise AND between the original grayscale image and its binary mask, we extract the pixel values corresponding solely to organoids. Conversely, by inverting the binary mask and repeating the operation, we isolate the background intensities. This enables us to compute separate histograms and assess the noise characteristics and overlap between classes.

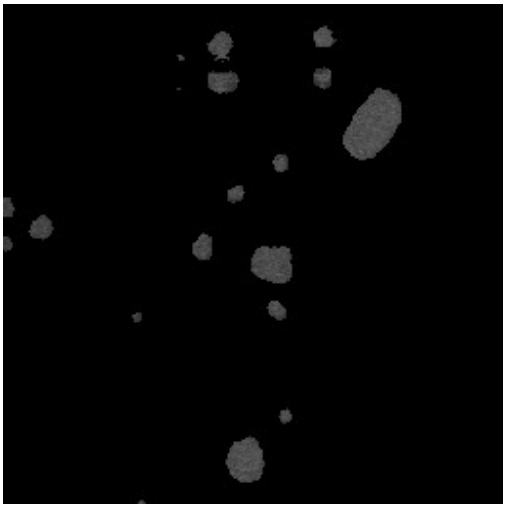

Fig.4 : Organoids only : the image is the result of a bitwise operation of type "AND" with Ground-truth image as a mask

Our initial observation is that both organoid and background intensities approximately follow Gaussian distributions, with a significant overlap due to noise and low contrast. To reduce this noise and smooth spatial discontinuities, we apply a Gaussian blur filter with a kernel size of 7×7. This operation suppresses high-frequency noise and groups together neighboring pixel values, making it easier to estimate global intensity trends.

While Gaussian filtering helps stabilize intensity distributions, it also causes blurring of fine edges, which is problematic given that organoid boundaries are already poorly defined. Nevertheless, this trade-off is necessary to enhance the separability of distributions in the subsequent thresholding step.

B. Statistical Thresholding and Image Cleaning

After denoising, we seek to define a threshold that retains as many organoid pixels as possible while eliminating a significant portion of the background. Based on the intensity histograms obtained post-Gaussian filtering, we choose a conservative intensity threshold of 59. This value corresponds to the 5th percentile of the organoid pixel distribution meaning we retain 95% of all organoid-related pixels.

Interestingly, applying this threshold also removes approximately 35% of background pixels, which fall below this cutoff. This gives us a strategic trade-off: we slightly truncate the organoid regions to significantly reduce background interference. We generate a new binary mask by thresholding the blurred image, then apply it to the original grayscale image via another bitwise operation. This yields a cleaner version of the image, where a substantial portion of the low-intensity background has been eliminated.

Fig 5. Identified Background as a binary mask to be removed from the original image

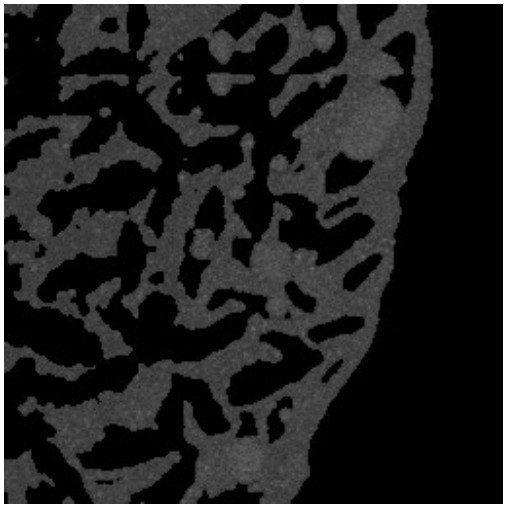

Fig. 6 : Original image cleaned thanks to the previous binary mask

C. Yen Thresholding for Segmentation

Having reduced the influence of noise and background, we apply Yen's thresholding method to the cleaned image. Yen's method is an entropy-based thresholding technique optimized for separating foreground and background in images with overlapping or skewed intensity distributions. Compared to Otsu's method, which assumes bimodal histograms, Yen's approach better handles the complex distributions encountered in our tomographic slices.

Yen's thresholding outputs a new binary mask, which we use as a preliminary segmentation of the organoids. This mask is later evaluated against the human-annotated ground truth to compute performance metrics such as the F1-score.

Among the many thresholding methods available, we compared three widely used approaches: Otsu, Li, and Yen. Otsu's method assumes a bimodal histogram, which rarely holds in our case, as the pixel intensity distributions of organoids and background often overlap significantly due to noise and low contrast. This often results in suboptimal thresholds and the loss of weak or diffuse organoid structures. Li's method, on the other hand, is based on cross-entropy minimization and tends to favor large homogeneous regions—leading to poor performance when segmenting small or low-contrast organoids.

In contrast, Yen's method maximizes a specific form of histogram-based entropy that does not assume bimodality and is more robust to multi-modal and overlapping distributions. This makes it particularly well-suited to our grayscale tomographic slices, where noise and structural complexity demand a more adaptive thresholding criterion. For completeness, Yen's method computes the optimal threshold that maximizes the sum of background and foreground entropies.

This concludes the first axis of our methodology—intensity-based segmentation—which provides a baseline understanding of how far one can go using classical intensity filtering and statistical thresholding alone.

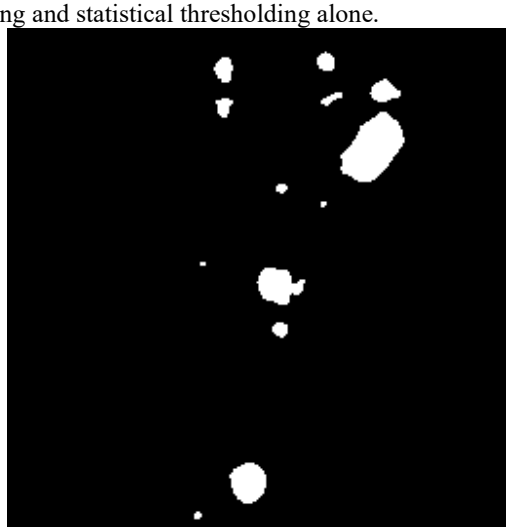

Fig 7. Processed image after Yen Thresholding and median blur filter

D. Edge-Based Segmentation and Elliptical Shape Filtering

While intensity-based thresholding provides a solid baseline, it struggles to distinguish tightly packed organoids and those with low contrast. To address this, we explored a second strategy based on edge detection and morphological analysis, aimed at capturing organoid contours more precisely and enforcing geometric constraints such as approximate ellipticity.

1) Preprocessing: Smoothing with Gaussian and Median Filters

Before detecting edges, we applied a sequence of denoising filters to enhance the visibility of structural boundaries while suppressing noise. The pipeline began with a Gaussian blur (kernel size 7×7) to smooth high-frequency noise, followed by a median blur to preserve edge discontinuities while removing isolated pixel-level noise (e.g., salt-and-pepper artifacts). A final Gaussian blur step was reapplied to homogenize the regions and reduce granularity. This sequence of operations helps balance edge clarity and robustness.

2) Edge Detection with Canny

After smoothing, we applied the Canny edge detection algorithm with both lower and upper thresholds set to 0. This configuration allows us to capture all gradient transitions in the image, producing a maximal set of edge candidates. While aggressive, this approach is useful in our setting where boundary contrast is extremely low and many true edges would otherwise be ignored.

3) Morphological Cleaning

To clean the binary edge map, we first performed morphological opening using a disk-shaped structuring element of radius 5. This operation removes small noise artifacts and breaks weak connections between adjacent objects. We also applied a hole-filling step to remove small internal gaps in closed edge contours. This post-processing prepares the image for shape analysis by ensuring the extracted regions are compact and topologically clean.

4) Elliptical Shape Test

Following edge extraction, we implemented an elliptical shape filtering step to discard regions that do not match the expected morphology of organoids. The assumption here is that most organoids exhibit approximately elliptical or ovoid shapes, despite their variability. For each detected contour, we fitted an ellipse and computed a set of geometric descriptors:

- Aspect Ratio Max: the ratio between the major and minor axes of the fitted ellipse (filters out elongated artifacts).
- Circularity Min: used to exclude highly irregular or fragmented shapes.
- Solidity Threshold: the ratio of the contour area to its convex hull area, helping to detect concavities.
- Size Filtering: to eliminate noise, we excluded components smaller than a predefined area threshold.

While this approach shows promise in isolating clean, ellipsoidal structures, its effectiveness is limited in practice. Due to the weak and noisy edges in many tomographic slices, contour continuity is often broken, and elliptical fits become unreliable. We also experimented with vesselness filters and gradient-based refinements, but none yielded consistently robust results across the dataset. As a result, edge-based segmentation remains a supplementary strategy, suitable for validation and qualitative insights but less reliable than intensity-based methods in this context.

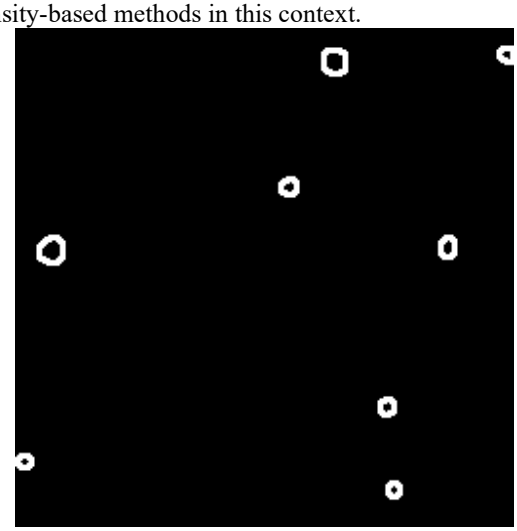

Fig. 8 : Image processed after Elliptical Shape Test. A few organoids are detected

E. Intensity Normalization and Gamma Correction: A Counter-Intuitive Strategy

To further improve segmentation, particularly in cases where contrast is too weak for traditional thresholding or edge detection, we investigated a less conventional preprocessing pipeline involving Z-score normalization and gamma correction. While these techniques are not commonly used together in biomedical segmentation workflows, their combination yielded surprisingly effective results in our experiments.

1) Z-Score Normalization

Z-score normalization standardizes pixel intensity values by subtracting the mean and dividing by the standard deviation of the image.

This transformation centers the intensity distribution and scales it, effectively reducing the influence of extreme values and enhancing the contrast between subtle tissue variations. In our case, this was particularly useful for isolating organoid structures embedded in low-contrast hydrogel environments.

2) Gamma Correction

Following normalization, we applied a gamma correction transformation. We experimented with several values of gamma, and found that values below 1 (e.g.,

significantly improved the separation between organoids and background. Gamma correction compresses high-intensity values and stretches low-intensity ones, thereby making faint organoid regions more prominent. Whatsmore, negative values makes it quite interesting, setting pixels values associated with organoids either really high or close to zero, improving their visibility to the human eyes.

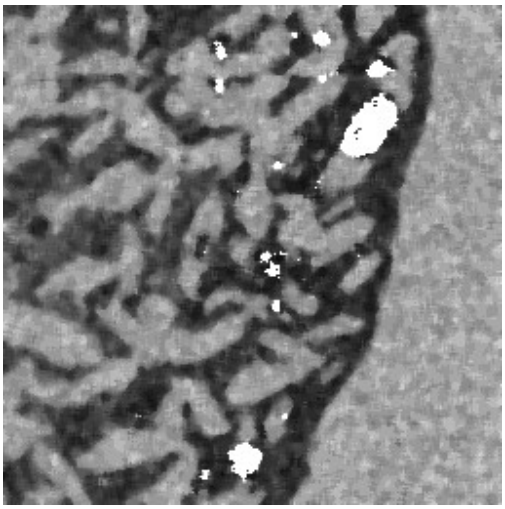

Fig. 9 : Processed image after Gamma Correction and Z-score Normalization.

3) Why This Works

This preprocessing strategy is counter-intuitive because Z-score normalization can flatten contrast, and gamma correction is usually used for visualization. However, in our application, the two operations complement each other: normalization regularizes the statistical scale across images, while gamma correction redistributes intensity values to amplify relevant structures. The resulting images are more amenable to fixed thresholding methods and lead to improved F1-scores during evaluation. Organoids are also brighter in general, as their mean are up to 10 pixels higher on average, which explains why the z-score normalization sets the organoid's pixels values higher.

This approach proved especially beneficial for organoids that were not clearly visible in the original grayscale images. It also performed well on early-stage organoids with low mass and poorly defined boundaries, highlighting its potential as a preprocessing module for segmentation pipelines — even beyond this specific application.

F. Evaluation Metrics: F1-Score and Comparison with MedSAM

$$\text{F1-score} = \frac{2 \times (\text{Precision} \times \text{Recall})}{\text{Precision} + \text{Recall}}$$

To quantitatively evaluate the quality of our segmentation approaches, we relied on the F1-score, a harmonic mean of precision and recall, computed between our predicted binary masks and the ground-truth annotations.

where:

- Precision is the proportion of correctly predicted organoid pixels among all predicted organoid pixels.
- Recall is the proportion of correctly predicted organoid pixels among all true organoid pixels.

This metric is well-suited to our task because it penalizes both false positives (e.g., hydrogel misclassified as organoid) and false negatives (e.g., missing weakly visible organoids), balancing the trade-off between over- and under-segmentation.

*1) **Evaluation of the Intensity-Based Approach***

Using the Gaussian-blur + Yen thresholding pipeline, we generated segmentation masks for each grayscale tomographic slice and compared them to the manually annotated ground-truth binary masks. The results yielded an average F1-score of 0.72, indicating reasonable overlap but highlighting the difficulty of detecting low-contrast organoids or separating overlapping ones.

Despite its simplicity, this method performed surprisingly well on well-isolated and medium-to-large organoids, especially when background noise was sufficiently reduced by preprocessing. However, smaller or poorly contrasted organoids were often missed or only partially segmented, contributing to lower recall values.

*2) **Baseline Comparison with MedSAM***

As a point of comparison, we employed MedSAM, a segmentation model based on the Segment Anything Model (SAM) adapted for biomedical imagery. Given a prompt (in this case, a center point or bounding box around the organoid), MedSAM generates a segmentation mask using a transformer-based architecture pre-trained on large-scale datasets.

Without any fine-tuning, MedSAM achieved an average F1-score of 0.78 on our dataset — outperforming our handcrafted intensity-based approach by 6 percentage points. MedSAM was particularly effective at detecting faint organoids and delineating complex boundaries, thanks to its ability to generalize across heterogeneous structures.

However, MedSAM also produced false positives in areas with artifacts or ambiguous textures, suggesting that some domain adaptation is still necessary for optimal performance.

G. Fine-Tuning MedSAM with Image Processing Techniques

Although MedSAM demonstrated strong performance in segmenting organoids, we sought to further optimize its performance through image preprocessing and prompt engineering. Since MedSAM is a general model, adapting it to the unique challenges of organoid segmentation required targeted adjustments.

1) Preprocessing Inputs for MedSAM

We first applied the same preprocessing techniques used for intensity-based methods (e.g., Gaussian blur, Z-score normalization, and gamma correction) to the inputs fed into MedSAM. These transformations enhanced the contrast between organoids and their hydrogel environment, making the organoid boundaries more prominent and improving MedSAM's segmentation accuracy. Specifically, we observed the following improvements: Z-score normalization ensured consistent pixel intensity scaling across images, reducing variability introduced by different lighting conditions in the tomography scans.

Gamma correction improved visibility of low-contrast organoids, which MedSAM could not segment accurately in the raw images.

By feeding these preprocessed images to MedSAM, we hoped to provide clearer, more uniform data, thus enabling the model to make more precise predictions.

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Algorithm	score
raw	0.7832071576049553
gaussian_blur	0.612684883429431
median_blur	0.5921579087756735
bilateral_filter	0.5137759118341643
clahe	0.4280442804428044
histogram_equalization	0.24139130434782607
sobel_edge	0.01288244766505636
canny	0.04020908725371934
laplacian	0.14822695035460992
unsharp_mask	0.7497661365762395
adaptive_threshold	0.13432282003710574

Fig. 10 : Evaluating finetuned MedSAM's F1-score on a pre-trained image processed with a computer vision algorithm

2) Designing Better Prompts

While MedSAM is versatile in segmenting general objects, fine-tuning its ability to detect specific organoid shapes required us to refine the input prompts. In our case, the prompt to the model was crucial for guiding its segmentation process. We experimented with several prompt strategies:

Point-based prompts: Providing a single point within each organoid allowed MedSAM to learn the center and expand its region of interest from that point. This strategy was particularly effective when organoids were well-separated.

Bounding-box prompts: In cases of closely packed or overlapping organoids, we provided a rough bounding box around the region of interest, which helped MedSAM to differentiate between multiple organoids within the same field.

Mask-based prompts: For images with clear but challenging organoid shapes, we used manually created binary masks as initialization, guiding MedSAM to refine its segmentation.

These prompt strategies, combined with image preprocessing, enhanced MedSAM's ability to segment organoids in diverse image conditions.

3) Results After Fine-Tuning

After fine-tuning with image processing techniques and optimized prompts, we tested MedSAM on the validation set of our dataset. The average F1-score improved to 0.83, a 5% increase over the baseline performance (0.78). This enhancement reflects MedSAM's improved capacity to segment smaller organoids, reduce false positives, and better capture organoid boundaries.

However, the model's performance was not flawless. While the preprocessing and prompt adjustments improved segmentation quality, MedSAM still struggled with highly irregular or partially obscured organoids. In such cases, the performance gains were less pronounced.

H. Conclusion

In this study, we proposed a comprehensive approach to the segmentation of organoids in grayscale tomographic images. Our methodology combined intensity-based techniques, edge-based approaches, and deep learning models to address the challenges posed by noisy, low-contrast images. Specifically:

Intensity-based thresholding, particularly using Yen's method, was effective for detecting well-contrasted organoids but struggled with poorly defined boundaries.

Edge-based methods, like Canny edge detection and elliptical filtering, provided additional insight into organoid shapes but were limited by weak boundaries in many images.

Z-score normalization and gamma correction offered a counter-intuitive but effective preprocessing pipeline that significantly improved the clarity of organoid regions, making them more amenable to segmentation.

Fine-tuning MedSAM with image preprocessing and improved prompting led to a noticeable increase in F1-score, particularly in terms of capturing smaller organoids and refining segment boundaries.

The combination of these techniques demonstrates the potential for hybridizing traditional image processing with deep learning models to solve complex segmentation tasks in biological imaging. While further research and model adaptation are needed, these findings highlight the importance of both image preprocessing and model fine-tuning in improving the robustness and accuracy of organoid segmentation.

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